

Electrical Characterization of Organic FETs

Temperature analysis for gvdP method

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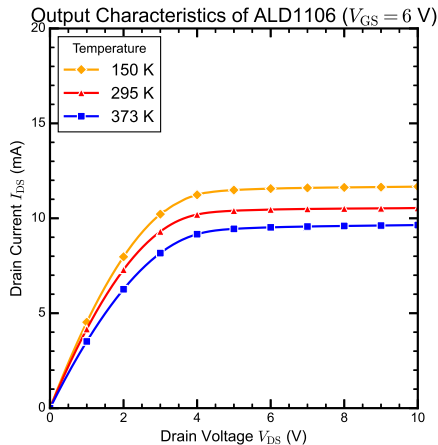
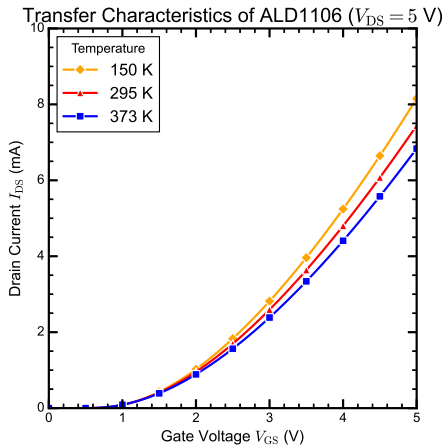


Figure: Transfer and Output Characteristics for different temperatures.

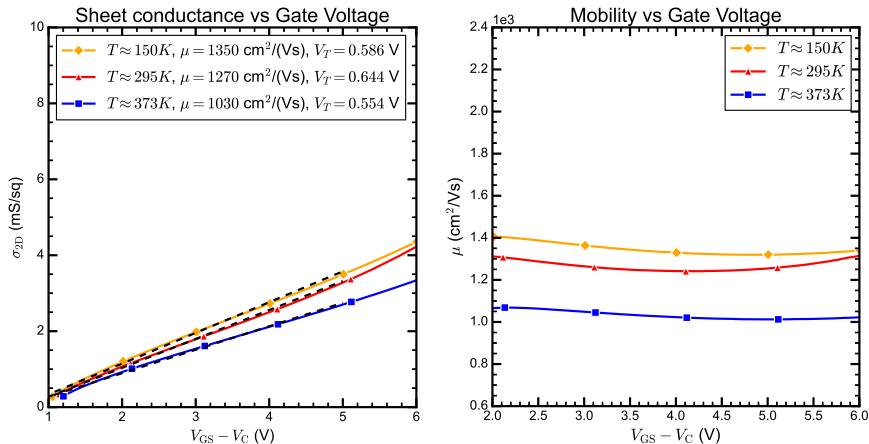


Figure: gvdP Method for ALD1106 for different temperatures.

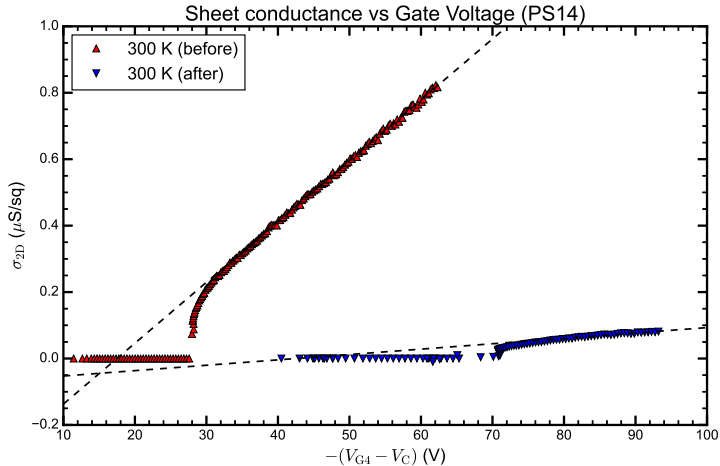


Figure: Conductance of PS14.

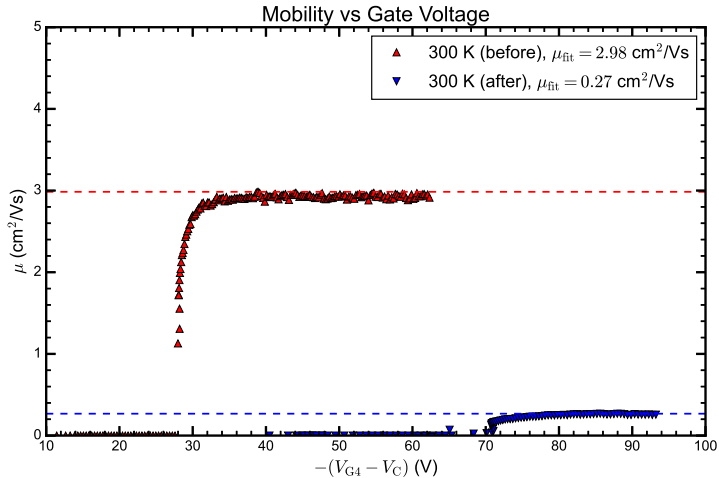


Figure: Mobility of PS14.

Main observation:

$$300\text{ K} \rightarrow 100\text{ K} \rightarrow 300\text{ K}$$

The thermal cycle causes a persistent change in device behavior:

$$\mu_{\text{fit}} : 2.98 \rightarrow 0.27\text{ cm}^2/\text{Vs}, \quad \Delta V_T \approx 40\text{ V}$$

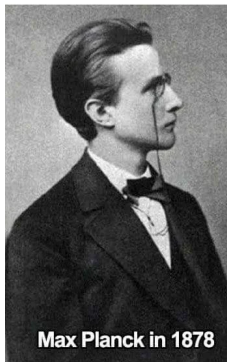
Interpretation:

The incomplete recovery indicates an irreversible or quasi-irreversible process.

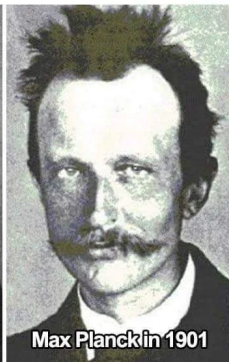
Possible origins are trapped charges, contact degradation, or reactions with residual water/oxygen.

These effects permanently modify the interface and reduce the apparent mobility.

What Quantum Physics does to a man



Max Planck in 1878



Max Planck in 1901

Thank you for your attention!
Any questions?